

Extinction is good—it solves future generations of non-human suffering and stops animal torturing
tochO'Brien '24 IGary; January 19; Research Assistant Professor at Lingnan University, fellow at the Hong
Kong Catastrophic Risk Centre, Research Associate at the Institute of Ethics In Technology, Ph.D. from the
University of Oxford Uehiro Centre for Practical Ethics; Brill, “The Case for Animal-Inclusive
Longtermism,” brill.com/view/journals/jmp/aop/article-10.1163-17455243-20234296/article-10.1163-17455243-20234296.xml DOI 7/181
azhao recutit also seems likely that the Big Future promise applies at
least as well to animals as it does to humans. At present there are around 8 billion human beings on the
planet, but we are outnumbered by other sentient vertebrate animals of whom there are at the very
least 10 trillion (10^{13}) and perhaps more than a quadrillion (10^{15}) (Tomasik, 2019). On conservative
estimates, then, animals outnumber human beings by between a thousand and a hundred thousand to
one, but the true population of animals may be orders of magnitude higher. Bristlemouth fish alone may
number in the quadrillions (Woods Hole Oceanographic Institution, 2019). This means that most sentient
individuals are nonhuman animals. If we assume that animal and human populations will remain at the
same ratio in the long term, then the long-term future has the potential to contain far more animal
suffering and wellbeing than human suffering and wellbeing. If we are impartially concerned with
maximizing wellbeing and minimizing suffering, then we ought to be very concerned about the long-term
future of nonhuman animals on the assumption that they will continue to constitute the vast majority of
welfare subjects. If Big Future inclines us to take longtermism for humans seriously, then it should also
incline us to take longtermism for animals seriously. It might be objected here that it is unlikely that
animal populations will remain at their current levels. Animal populations might decline drastically and
remain at low levels for the rest of our planet’s period of habitability. In fact, it seems that wild animal
populations have decreased dramatically in the past 50 years, as human development has reduced the
habitats for many species, and this process might continue until relatively few wild animals remain. A
2020 report by the WWF suggests that there has been a 68% decline in the average populations of
monitored vertebrate species since 1970. It must be borne in mind, however, that this report is
concerned specifically with biodiversity, not with animal populations generally, and the findings of the
report are compatible with global animal populations staying the same or increasing. For example, the
report mentions invasive species as a major cause of biodiversity loss (2020, 22, 54) but if invasive
species are replacing native ones this need not cause a net decrease in animal populations. Similarly,
reductions in the numbers of large animals can be more than compensated for by increases in the
numbers of small animals. Other causes of biodiversity loss more clearly seem to indicate general
population declines, however. Future humans might also eliminate domesticated animal populations,
for example by switching from animal agriculture to plant-based or synthetic alternatives. If this
happens then the Big Future promise is not true of animals, and so the longtermist thesis does not apply
to them. In response, we cannot simply assume that animal numbers will drop significantly and remain
low indefinitely. This will depend in large part on the values and actions of human beings over the
coming centuries, and these are yet to be determined. Even if one were somehow certain that humanity
will eventually drive all the other animals to near extinction, in order to do so humanity itself must
survive long enough. Ord (2020) puts the risk of human extinction within the next century at about 17%.
Newberry (2021, 4) cites several experts on existential risk who also put the risk at around 20% in the
next 100 years. If one accepts that there is a significant risk of human extinction in the next 100 years,
then in order to maintain that there is a high probability that humans will drive animal populations to
near extinction one would also have to believe that they will do so relatively soon, and that animal
populations will never rebound after human extinction. Secondly, as we will see in Section 3, there are

some ways in which humanity could ensure a long future for animals even if animals do eventually go extinct on Earth. Thirdly, if one believes that animals generally live good lives, or that they could live good lives with human assistance, then one ought to be concerned about animal extinction too, as this might reduce total wellbeing in the long term.¹⁵ A more fundamental point is that longtermists generally think about the size of the future in terms of expected population, rather than in terms of the most likely outcome or the median across various possible outcomes.¹⁶ In essence this means that they generate high and low estimates for possible future populations then weigh them according to their probabilities thereby establishing a figure for the expected population. They find that, since human populations have the potential to be truly vast, the resulting figure is also vast, even when they account for the possibility of very low populations or total extinction. As Greaves and MacAskill say: Even a 50% credence that the number of future beings will be zero would decrease the expected number by only a factor of two. In contrast, a credence as small as 1% that the future will contain, for example, 1 trillion beings per century for 100 million years ... increases the expected number by a factor of 100. The same consideration applies to the animal population – since it has the potential to be very large, we would need to have an unreasonably high degree of confidence that their population would decline to zero or near-zero in order to endorse a low figure for their expected population. But, if a high expected human population is sufficient to ground longtermist concern for human beings, then a high expected population of animals ought to be sufficient to ground longtermist concern for them too. There is one final consideration relevant to the application of the Big Future premise to animals. Longtermists often bolster their case by arguing that human beings might one day colonize other planets, or that we might one day create human minds on computers, all of which would multiply the possible future population enormously.¹⁷ If we consider such huge expansions to the human population likely, but think it very unlikely that future animal populations will be expanded in the same way, then we might be justified in thinking that the vast majority of future sentient beings will be humans (or digital human minds) and so it is much more important to ensure that the long-term future goes well for them than for animals.¹⁸ That is, though the absolute number of animals in the future remains large, it would be dwarfed by the human population, and thus rendered irrelevant to our longtermist decision-making. This is of course highly speculative. There is no reason in principle however why these technologies could not also multiply animal populations. Tomasik (2018) speculates about the possibility that future human beings might choose to create digital worlds replete with sentient digital animals, perhaps in natural settings similar to those on Earth in which they might suffer greatly. Future humans might also choose to create digital minds in order to maximize hedonic wellbeing, and since animal minds are so much simpler than human minds they might do so by creating digital animal minds rather than digital human minds. A human brain contains around 86 billion neurons, while a mouse brain contains a mere 70 million. Assuming that the difficulty of creating a digital mind increases proportionally with the number of neurons required, then a utility-maximizing would-be digital mind creator might be advised to create digital mice rather than digital people. In Section 3 I will consider the possibility that humans might one day spread wild and/or domesticated animals to other worlds. Here I will only say that this possibility is not negligible, and in fact there are some ways in which it is much easier to spread animal life than human life, and so we cannot discount the possibility that animal populations will also be multiplied across physical and digital space. It is possible therefore that animal populations will be multiplied by these technologies just as human populations might be. Even if we think this unlikely, so long as we do not think it impossible, the huge numbers involved serve to increase the expected future animal population significantly.

Of course, it is extremely difficult to make accurate predictions about far future population sizes for humans or animals. There are so many known and unknown factors that could cause the population to increase dramatically, fall to zero, or anything in between. This uncertainty may count against the plausibility of longtermism generally but is not an argument against including animals in the longtermist framework. To show that Big Future is true of humans but not of animals it is not enough to point to the uncertainty about future population sizes. Rather, it must be shown that there is an asymmetry between humans and animals which means that, while the future human population is large in expectation, the expected future animal population is not. I have argued above that there is no such asymmetry – if we believe that the expected human population is large, we should think the same about the expected animal population. Furthermore, there are some factors which give us reason to expect a large animal population rather than a large human one. For example, humans are just one species among many. We are an exceptional species, and it is likely that we have the means to survive threats to our existence which other animals do not, and so perhaps we are less likely to go extinct than other species. However, there are millions of species on the planet. Though our likelihood of going extinct might be lower than that of any other species, it is unlikely that our chances of survival are greater than those of all other species on Earth combined. It may be the case then that humans go extinct while other animals live on indefinitely, thus ensuring Big Future for animals, but not for us. If it is unwarranted to be confident about the size of the future human population, it is doubly so to be confident that our population will increase and the population of animals will shrink or vanish. Finally, there are two considerations that indicate that longtermism is particularly important for animals. The first concerns our current capacity to help wild animals, which is far more limited than it is likely to be in the future if we act today to ensure that we expand that capacity. It is presently possible to help wild animals in limited ways by vaccinating them against some diseases, leaving out food supplements in times of extreme scarcity, and rescuing and rehabilitating them when they are affected by natural catastrophes. The main causes of wild animal suffering, however, are systemic features of the natural world, primarily reproductive strategies, endemic scarcity, and various forms of antagonism, including predation, which we currently have little ability to influence. In order to seriously improve the lives of wild animals we need to develop our scientific knowledge of the causes of suffering and wellbeing in diverse species of wild animals, we need to understand how the various components of the biosphere interact so that we can better predict the results of our interventions, and we need to develop the technology to intervene safely and effectively. These developments may take decades or centuries to come to fruition. Those who are concerned with animal wellbeing might therefore think that it is important to begin the research work that will allow us to significantly help wild animals in the future. The second consideration concerns the relative stakes for humans and animals in the future. The bad futures for humans considered by longtermists often involve premature extinction, or failure to reach full technological maturity. Futures like these are only 'bad' compared to the positive futures which we could have, but they are not intrinsically bad. Of course, there are possible futures in which human beings suffer significantly, for example if a sufficiently bad global totalitarian regime took over, or if a super-intelligent AI came to positively value human suffering, but these futures are relatively unlikely. The most likely futures open to animals on the other hand are intrinsically bad, in the sense that they involve large amounts of suffering for large numbers of animals over very long periods. Unlike humans, who have the intelligence and technological skill to solve the problems facing them, animals seem almost helpless against the natural forces that cause them to suffer. We might say that the default state of animals is to suffer. Since the most probable long-term futures are extremely bad for animals, we have

additional reason to take measures today to try to change our trajectory to avoid those futures. 3 Causal Efficacy

In this section I will argue that Causal Efficacy applies to animals. By utilizing the longtermist concepts of persistent states and value lock-in, it is possible to identify some actions that we can take today that we can reasonably expect to make the long-term future go better or worse for animals. First, I describe was as a negative-value persistent state which, absent human intervention, we can expect to last for a very long time. I also describe some ways in which human agency could increase the scale and persistence of was. Second, I argue that there is a danger that humanity might lock in values which will make the long-term future much worse for animals by spreading and perpetuating animal exploitation, and by making future people less concerned about mitigating was. Furthermore, if it is the case that humanity will soon expand into space, then this is a very strategically important time at which to avoid locking in bad values.

3.1 was as a Negative-Value Persistent State On Earth trillions of animals suffer and die painfully each year as a result of purely natural processes, while a much smaller number seem to live good lives. It is likely that the vast majority of lives lived by wild animals today are bad, and plausible that aggregate welfare across all lives on Earth is also negative.¹⁹ Furthermore, this does not seem to be an accidental feature of life on our planet; rather it is caused by structural features of evolution by natural selection, and so was is likely to continue to dominate animal wellbeing for as long as life on Earth continues to be governed by those processes. In fact, since these natural harms are the predictable result of natural selection, it is reasonable to think that similar outcomes will obtain on any planet on which sentient life evolves.²⁰ It is difficult to quantify just how bad was on Earth is, but we can make some reasonable estimates. Extremely conservatively we can say that ten trillion animals die each year after enduring only 1 minute of not suffering. This would amount to approximately 18.2 million years of experienced suffering each year across all animals on Earth. If this does not sound that bad, remember that the degree of suffering that we are talking about is high – for example, being eaten alive or dying of starvation. This estimate is highly conservative in many ways. First, it takes the lowest estimate of wild animal populations. Second, it falsely assumes that the number of deaths per year is equal to the number of animals alive at any given time, but since the vast majority of animals are small short-lived ones who breed frequently and in vast numbers, the number of deaths per year must be much higher than this. Third, it assumes only one minute of not suffering. This may be true of animals such as small fish who are quickly eaten by predators immediately after hatching, but it must be remembered that many animals also starve, or die from disease, and even those who are killed by predators must sometimes endure more than a minute of suffering from being pursued and eaten more slowly. A reasonable higher estimate would be that there are a quadrillion deaths per year. If we assume again only one minute of not suffering each, that comes to 1.82 billion years of experienced suffering per year. R-selection makes suffering predominate

Sittler-Adamczowski '16 [Thomas; December; University of Oxford; Journal of Practical Ethics, "Consistent Vegetarianism and the Suffering of Wild Animals," <http://www.jpe.ox.ac.uk/papers/consistent-vegetarianism-and-the-suffering-of-wild-animals/> DOI 7/18 Thomas M. Sittler-Adamczowski is an academic researcher and author specializing in ethics and artificial intelligence forecasting, best known for his award-winning work in practical ethics] azhao recut If animals like free-range cows have lives that are not worth living, almost all wild animals could plausibly be thought to also have lives that are worse than non-existence. Nature is often romanticised as a well-balanced idyll, so this may seem counter-intuitive. But extreme forms of suffering like starvation, dehydration, or being eaten alive by a predator are much more common in wild animals than farm animals. Crocodiles and hyenas disembowel their prey before killing them (Tomasik 2009). In

birds, diseases like avian salmonellosis produce excruciating symptoms in the final days of life, such as depression, shivering, loss of appetite, and just before death, blindness, incoordination, staggering, tremor and convulsions (Michigan Department of Natural Resources). While a farmed animal like a free-range cow has to endure some confinement and a premature and potentially painful death (stunning sometimes fails), a wild animal may suffer comparable experiences, such as surviving a cold winter or having to fear predators, while additionally undergoing the aforementioned extreme suffering (Tomasik 2013). Wild animals do experience significant pleasure, for instance when they eat, play, have sex, or engage in other normal physical activity. One reason to suspect that on average this pleasure is outweighed by suffering is that most species use the reproductive strategy of r-selection, which means that the overwhelming majority of their offspring starve or are eaten shortly after birth and only very few reach reproductive age (Morta 2010; Ng 1995). For instance, 'In her lifetime a lioness might have 20 cubs; a pigeon, 150 chicks; a mouse, 1000 kits' (Hapgood 1979), the vast majority of which will die before they could have had many pleasurable experiences. Overall, it seems plausible that wild animals have worse lives than, say, free-range cows. If vegetarians think it's better for the latter not to exist, they must believe the same thing about wild animals. A second important empirical fact is that wild animals far outnumber farmed animals. Using figures from the FAO, Tomasik estimates that the global livestock population is 24 billion (including 17 billion chicken) (Tomasik 2014). I restrict my count of wild animals to those at least as complex as chicken or small fish, which vegetarians clearly believe do have moral weight. Using studies of animal density in different biomes, Tomasik estimates conservatively that there are at least 6×10^{10} (60 billion) land birds, 10^{11} (600 billion) land mammals, and 10^{13} (60 trillion) fish. Animals in each of these categories alone are several times more numerous than livestock. If wild animals' well-being is indeed below the threshold for a life worth living, and the above numbers are remotely correct, the scale of wild animal suffering is vast. As Richard Dawkins writes, 'During the minute it takes me to compose this sentence, thousands of animals are being eaten alive; others are running for their lives, whimpering with fear; others are being slowly devoured from within by rasping parasites; thousands of all kinds are dying of starvation, thirst and disease.' (Dawkins 1996) If they accept the premises so far, consistent vegetarians should focus on preventing the existence of as many wild animals as possible, since even a small reduction in the global number of wild animals would outweigh the impact of ending all livestock production. For example, they could reduce animal populations by sterilising them, or by destroying highly dense animal habitats such as rainforests. It may even be the case that vegetarians should react to this argument by eating more meat, since feeding livestock requires more surface area for agriculture, and fields contain far fewer wild animals per square kilometre than other biomes such as forests (Mathony and Chan 2005, 585). Of course, to the extent that it is more difficult to reduce wild animal populations than farm animal populations, vegetarians should focus more resources on the latter. But it seems implausible that it would be over a hundred times more difficult to achieve the same proportional reduction, which is what would be needed to reverse my conclusion that wild animal suffering dominates. There could be some simple ways, for instance, for vegetarians to reduce habitat sizes: supporting the construction of large parking lots, or donating to a pro-deforestation lobby. In the final paragraph, I touch upon the issue of how most effectively to reduce wild animal suffering.

Non-human suffering is the largest impact, humans are cognitively biased against animal pain. Any uncertainty means you err [SIDE]

Moore '16 (Ole Martin; 2016; University of Oslo, Centre for the Study of Mind in Nature; Nord J Appl Ethics, "ethics of wild animal suffering."

<http://www.olomartinmoon.com/wp-content/uploads/TheEthicsofWildAnimalSuffering.pdf> Ole Martin Moon is a Norwegian philosopher who works primarily with applied ethics and value theory.¹ Although solipsism at the species level might make sense within religious contexts where humans are taken to have originated separately from all other animals, it coheres well with neither neuroscience nor evolution. Comparing ourselves to sparrows, rabbits, and bears, we may observe that we have the same kind of neurons, the same main brain parts, and the same pain pathways (C and A delta fibers) that they have. Sparrows, rabbits, and bears, moreover, react to noxious stimuli the same way we do, and they stop doing so when anesthetized (see Griffin & Speck 2004; Dawkins 2015). Since we and other animals are genetically, neurologically, and functionally very close, we would need weighty evidence to conclude that, despite these similarities, humans work in fundamentally different ways from other animals: humans consciously, animals non-consciously.¹

<<CONDENSED, NONE OMITTED>> Increased understanding of animal helped spur the animal ethics movement. Keeping animals in small cages, castrating them without anesthetics, and branding them with glowing irons—practices that, if performed on humans, would land the perpetrator in prison for decades—are common farming practices around the world. Millions of farm animals live and die under such conditions. Opposing human disregard for animal welfare, Peter Singer (1990) famously argues that just as we have gradually expanded our circle of moral concern to encompass ethnic groups other than our own, and finally humanity as a whole, we should further expand it to include other sentient species. According to Singer, it is suffering as such that is bad, and it is bad whoever experiences it. Though the animal ethics movement is commendable, its circle of moral concern has hitherto expanded almost exclusively to captive animals. With very few exceptions—most notably, David Pearce and Jeff McMahan, whom I shall discuss in detail below—animal ethicists have failed to adequately take into account the suffering of animals living in the wild. Wild animals, however, vastly outnumber captive animals, and arguably, billions of wild animals live lives that are even more painful and distressing than those of their captive counterparts. Though it might well be difficult to alleviate suffering in the wild, and comparatively easier to alleviate suffering caused by humans, disregarding wild animal suffering from the outset involves a form of anthropocentrism that, sadly, enjoys wide acceptance even among those who purport to oppose the doctrine. We might dub this the second anthropocentrism. While traditional anthropocentrists are concerned only with human suffering, anthropocentrists of the second kind are concerned only with human-caused suffering. I will suggest, however, that if we take suffering as such to be bad (roughly along the lines that Singer does), it is unclear why the species membership of those who cause the suffering is morally relevant while the species membership of those who suffer is not. My aim in this paper is not to sway those who are indifferent to animal welfare. Rather, my aim is to make those who are concerned with animal welfare more concerned with the welfare of wild animals. Moreover, I shall exclusively discuss welfarist concerns, so if there are other grounds to care for animals, they lie beyond the scope of this paper. My discussion is limited to mammals and birds, the reason for which is that these are the animals whose ability to suffer is least disputed. If fish, amphibians, reptiles, and/or invertebrates can also suffer, my conclusion is amplified. The empirical side Let me start by defending three empirical claims: (1) that there are vastly more wild than captive animals; (2) that wild animals have the same capacity to suffer as captive animals; and (3) that many, perhaps most, wild animals suffer at least as much as their captive counterparts. These are all empirical claims that say nothing about the value significance of wild animal suffering. As such, we should accept or reject these claims irrespective of our ethical views. How many captive animals are there? According to the Food and Agriculture Organization of the United Nations (2014), the total number of livestock in the world is—at

any given time—roughly 25 billion, the majority of which are chicken, followed by ducks, cattle, and sheep. Although this figure leaves out pets and laboratory animals, let us take for granted, for the sake of convenience, that the number of livestock is roughly representative of the number of captive animals. How many wild animals are there? According to Brian Tomasik's (2014a) estimations, which are generated from research data on the typical prevalence of various animals in various environments coupled with data on the global prevalence of those environments, there are—at any given time—between 60 and 200 billion birds and between 100 and 1,000 billion mammals. If we assume the middle estimate for both birds and mammals, there are, at any given time, 700 billion wild birds and wild mammals combined. This is roughly 25 times the number of birds and mammals in captivity. (If we were to include in our estimates fish, amphibians, reptiles, and invertebrates, which are rare in human captivity but very prevalent in the wild, we would end up with thousands of times more wild than captive animals.) A further empirical premise is that wild animals have the same ability to suffer as captive animals. By this I simply mean that if you tear the skin of both a wild and a captive animal, there is no compelling reason to believe that this would hurt more for the captive animal than for the wild animal. In fact, if we were to conclude that there is a difference between the two, we should probably conclude that while captive animals are more docile (due to drugs and lack of stimulation), wild animals remain sharp and focused. Let us assume, however, that the ability to suffer is the same, or roughly the same, in captive and wild animals. How much do wild animals actually suffer? Very likely, some wild animals suffer very little. Some live long and peaceful lives, have few natural enemies, and have ample supplies of food. When they die, moreover, many animals die quick and painless deaths. The fact that some lives in the wild are pleasant, however, does not contradict the fact for billions of wild animals, life is filled with suffering. One prominent source of suffering is predation. Every day, millions of animals are eaten alive, and though some of them are killed quickly, larger animals will often stay alive for minutes or hours before they die of blood loss, suffocation, drowning, or internal bleeding from poisoning (Tomasik 2014b). While some become paralyzed, and are likely to feel nothing, others feel excruciating pain. Predation is a very visible cause of suffering. In response to this, Tyler Cowen (2003) and Jeff McMahan (2010) have argued that if we can easily prevent a predator attack, we have at least a pro tanto moral reason to do so. In their view, the way predators kill their prey is often so gruesome that if a human were to treat animals similarly, we would have strong reasons to intervene – and for the animal that is eaten alive, the species membership of the attacking predator is likely to matter very little. Though this is an important observation, I think Cowen and McMahan fail to appreciate that suffering caused by predation is likely to account for only a small fraction of the total suffering in nature. Though death from predation might be the most violent and visible cause of suffering, deaths from disease and parasites tend to be more drawn out in time. The same is true of deaths from droughts, floods, and freezing. Life in the wild is also a constant quest for nutrition; at any given time, thousands of animals are in the process of starving to death. Though there is no agent responsible for this suffering, and though it might be hard for us to detect it, the suffering is nonetheless real and prevalent. When a parent animal starves or freezes to death, gets eaten, or dies from disease, its young offspring will often face an equally painful death. This borders on an important point, namely that most suffering in nature is likely to be endured by very young individuals. The reason is not primarily that many parent animals die (although that is also the case), but that most wild animals give birth to many more offspring than are likely to reach adulthood. While humans normally give birth to just one child per year, and provide extensive care to each child (this is called the K-selection strategy), many animals follow a different reproductive strategy: they give birth to dozens or hundreds of offspring every year, and care very little for each individual (the r-selection

strategy). These strategies both work to spread the parents' genes in the population, but the r-selection strategy—which is most common in smaller animals—leads to enormous amounts of suffering because of the very large number of young individuals that are left to starve to death or get eaten, either by their stronger siblings or by other predators (for an elaboration, see Norta 2010). If the average female in a given animal population gives birth to 50 offspring every year—and the population size remains stable year after year—then the majority of individuals in that population will be individuals dying before reaching adulthood. If we grant that animals become conscious shortly after birth, as we assume to be the case with humans, their deaths will often involve pain, and since their lives are very short, they will have very few good things in life to weigh up for all that is bad. For these reasons, Richard Dawkins is almost certainly correct when he writes: The total amount of suffering per year in the natural world is beyond all decent contemplation. During the minute that it takes me to compose this sentence, thousands of animals are being eaten alive, others are running for their lives, whimpering with fear, others are being slowly devoured from within by rasping parasites, thousands of all kinds are dying of starvation, thirst and disease (Dawkins 1995: 131-32). <<PARAGRAPH INTEGRITY RESUMES>> Wild animal suffering is mostly invisible to us. Humans never see the vast majority of wild animals, and those that are seen by us are predominantly healthy and moving. We do not see the young individuals starving to death or the adult individuals being devoured by parasites, and we must keep in mind that even if we saw them, their suffering would often not be apparent to us. While we have evolved to pick up pain cues from other human beings, we are much worse at picking up pain cues from non-human animals, especially those that are genetically remote from us. Moreover, many animals hide signs of weakness and disease to avoid attracting predators (including humans) looking for easy prey. When Thomas Hobbes wrote that life, in the state of nature, is “solitary, poor, nasty, brutish, and short,” he meant human life (Hobbes 1651/1996: XIII.9). It seems, however, that the description is also fitting for the lives of many non-human animals. Because of the brutality of wildlife, one could even make the provocative case that a typical life in the wild is even more painful and distressing than a typical life in human captivity. Although factory farming is often grotesque, animals in captivity are seldom killed in ways that draw out their deaths over several minutes or hours; they are not exposed to predators until they are slaughtered; they typically have access to sufficient amounts of food and water; and the temperature tends to be comfortable. Concerning larger animals, such as cattle, individuals with serious diseases will often be euthanized. For this reason, it is not clear that the average life in the wild is filled with any less suffering than the average life in captivity. However, even if wild animals do, on average, suffer less than captive animals, the sheer number of wild animals is still so overwhelming that the majority of suffering on Earth almost certainly takes place among animals living in wild nature. Prioritize species-neutral approaches to ethics—anything else is a slippery slope to genocide Jaeger '24 (Bastian; February; Researcher and Assistant Professor in the Department of Social Psychology at Tilburg University; Tilburg University, “Prejudice Across Species Lines: Testing the Link Between Human-Directed Prejudice and the Devaluation of Non-Human Animals,” pp. 3-10 Jaeger is a highly credible author specializing in ethics) Zhao recut Philosophers and scientists have noted the many similarities between the systemic exploitation of animals and (past and ongoing) systemic discrimination against certain human groups, such as racism and sexism (Dhont et al. 2020; Norta, 2010; Singer, 1975). Building on these observations, an emerging field of study is exploring the psychology underlying the differential treatment of beings based on their species membership (Caviola et al. 2019; Dhont et al. 2020). These attitudes and behaviors are often subsumed under the label of “speciesism”, a term that is meant to highlight parallels with racism and sexism (Norta, 2010; Singer, 1975). But parallels may not only exist at the

surface. Some have proposed that both prejudice against humans and prejudice against animals are the product of the same underlying psychology (Dhont et al., 2016; Dhont & Hodson, 2014). Thus, generalized prejudice—the observation that people who are prejudiced against one social group also tend to be prejudiced against other social groups—may not only generalize across different social groups (Akrami et al., 2011; Bergh & Brandt, 2023), but also across species lines. The present studies test this hypothesis, examining whether people who show prejudice against different social groups also show prejudice against animals. The studies are based on large, demographically diverse samples of participants (Study 1: 56,759 participants from 46 countries, Study 2: 1,566 participants from the Netherlands) and examine a host of attitudes, beliefs, emotional responses, and behaviors related to the devaluation of animals. Prejudice against humans and animals Decades of research have shown that people who derogate and devalue some groups also tend to devalue and derogate others, an individual difference that is referred to as generalized prejudice (Akrami et al., 2011; Allport, 1954; Dhont et al., 2016). One explanation for why different forms of prejudice often come as a package deal is that they are underpinned by the same beliefs and ideologies. Two traits in particular are often invoked to explain the causes of generalized prejudice: right-wing authoritarianism (RWA), a preference for conformity, traditional values, and deference to authority (Altemeyer, 1981), and social dominance orientation (SDO), a preference for group-based dominance hierarchies and social inequalities (Ho et al., 2015). Building on this work, recent theoretical models propose that generalized prejudice and the devaluation and derogation of animals—manifested, for example, in the common practices of meat consumption, animal experimentation, and recreational hunting—share the same underlying psychological mechanisms (Costello & Hodson, 2010, 2014; Dhont et al., 2016). SDO in particular has been highlighted as a proximate cause of prejudice against humans and animals alike (Dhont & Hodson, 2014; Sibley & Duckitt, 2008). According to the social dominance human-animal relations model (SD-HARM), preferences for group-based hierarchy and inequality, as captured by measures of SDO, motivated prejudice towards both humans and non-human animals (Dhont et al., 2016). Moreover, the interspecies model of prejudice posits that beliefs about the fundamental divide between humans and animals, and human superiority in particular, facilitate the derogation of human outgroups by likening them to lower-status animals (Costello & Hodson, 2010, 2014). There is considerable empirical support for the proposed links between prejudice against humans and animals. Individual differences in speciesism (i.e., attributions of lower moral worth to animals) are positively related to measures of racism and sexism (Caviola et al., 2019; Dhont et al., 2016), and broader antisocial personality traits such as low empathy, entitlement, and callousness (Hopwood & Blodorn, 2020). For example, a study by Jackson (2019) showed that participants who scored higher on speciesism also indicated more negative attitudes towards a host of social groups (measured with feeling thermometers), and this association was particularly strong for groups that are typically marginalized (e.g., homosexuals, Muslims, and disabled people). The killing and consumption of animals is perhaps the most typical symptom of the devaluation of animal lives. Evidence suggests that meat eaters are also more likely to be prejudiced against human groups (Vesor et al., 2015). Lay people may even be aware that generalized prejudice extends to non-human groups: Participants expected that a person with speciesist attitudes would be less supportive of gay, Black, and women's rights (Everett et al., 2019). Finally, there is evidence showing that interventions that tend to reduce prejudice towards humans groups, imagined contact with a member of the group, may also succeed in reducing prejudice towards animals (Auger & Amiot, 2019). There is also evidence suggesting that social dominance orientation can explain prejudice against both humans and animals. Decades of research have shown that SDO predicts prejudice against various social groups (Bergh &

Brandt, 2023; Ho et al., 2015; Sibbey & Duckitt, 2008). Recent work has found similar associations between SDO and prejudice against animals. People who score higher on SDO also score higher on speciesist attitudes (Caviola et al., 2019; Dhont et al., 2016; Dhont & Hodson, 2014; Graça et al., 2018; Hopwood & Blöndorn, 2020). They are more likely to think that hunting and fishing are justified (Hopwood & Blöndorn, 2020), more strongly endorse the use of animals by humans (Hyers, 2006), and are less likely to condemn acts of animal cruelty (Jarmakowski-Kostrzanowski & Radkiewicz, 2021). Several studies have shown that meat eaters score higher on SDO than vegetarians or vegans (Dhont & Hodson, 2014; Krings et al., 2021; Voser et al., 2015) and high-SDO individuals are less motivated to reduce their meat consumption because of animal welfare concerns (Hopwood & Blöndorn, 2020). Research by Dhont and colleagues (2014, 2016) directly tested whether SDO can account for the links between prejudice towards humans and animals. Across multiple studies, they found that speciesist attitudes were positively correlated with ethnic prejudice (i.e., more negative attitudes towards immigrants and various ethnic minorities). SDO was positively related to both and, crucially, the positive association between speciesism and ethnic prejudice was nonsignificant when controlling for SDO. These findings were replicated by Jackson (2019), who found positive associations between SDO, speciesism, and prejudice towards marginalized social groups (although there was still a significant link between speciesism and prejudice towards humans when controlling for SDO). Thus, accumulating evidence supports the notion that generalized prejudice also extends to animals and that individual differences in SDO can explain this link. <<CONDENSED, NONE OMITTED>> The Present Studies Prior work mostly supported the notion that human- and animal-directed prejudice are linked, but they also share a number of limitations. Previous studies overwhelmingly relied on participants from English-speaking countries who were recruited from university participant pools or online platforms, such as Amazon MTurk or Prolific (Auger & Amiot, 2019; Caviola et al., 2019; Dhont et al., 2016; Everett et al., 2019; Hopwood & Blöndorn, 2020; Hyers, 2006; Jackson, 2019). That is, evidence from demographically diverse samples outside of the Anglosphere is scarce, raising questions about the generalizability of results. To address this issue, the present studies leverage large, demographically diverse panel data sets (Study 1: 56,759 European participants, Study 2: 1,566 Dutch participants). Specifically, Study 1 relies on data from 46 countries, spanning Northern Europe (e.g., Sweden, Denmark, Iceland), Western Europe PREJUDICE ACROSS SPECIES LINES 6 (e.g., France, Ireland, Luxembourg), Eastern Europe (e.g., Russia, Poland, Azerbaijan), and Southern Europe (e.g., Greece, Turkey, Portugal). It is also unclear how robust the positive association between human-directed and animal-directed prejudice is when controlling for various socio-demographic factors. For example, men and conservatives often show higher levels of prejudice against marginalized groups and prejudice against animals (Caviola et al., 2019; Hodson & Dhont, 2015) and it seems like these associations are not only due to differences in SDO (Dhont & Hodson, 2014; Graça et al., 2018). The present studies explore (a) how various socio-demographic variables (i.e., gender, age, educational attainment, income, rural vs. urban living environment, religiosity, and political orientation) are related to attitudes, beliefs, emotions, and behaviors related to the devaluation of animals and (b) whether the link between human- and animal-directed prejudice still emerges when controlling for these variables. Finally, the majority of previous studies examined prejudice towards humans and animals by measuring general attitudes and beliefs towards these groups (e.g., Caviola et al., 2019; Dhont et al., 2016). The present studies rely on a larger and more diverse set of attitudes, beliefs, and emotional responses towards animals, including the endorsement of human supremacy beliefs (Study 1), beliefs about the ethicality and animal friendliness of animal agriculture, and feelings of shame and guilt when thinking about meat consumption (Study 2). Study 2 also examined specific

behavioral expressions of the devaluation of animals, including participants' frequency and quantity of meat consumption and their willingness to buy meat replacement products. Thus, the present studies test whether the link between human- and animal-directed prejudice generalizes across a conceptually diverse set of operationalizations of animal-directed prejudice. In sum, the present studies leverage large, demographically diverse panel data sets with different operationalizations of human- and animal-directed prejudice to provide a more severe test of the hypothesis that generalized prejudice also extends to animals. All data and analysis scripts are available at the Open Science Framework (<https://osf.io/dxrq5>). Study 1

Study 1 was a first test of the hypothesis that people who are more prejudiced towards human social groups are also prejudiced against non-humans. This hypothesis was tested with a large, demographically diverse, cross-national data set with 56,759 participants from 46 European countries. Specifically, associations between human-directed prejudice and beliefs that humans are meant to rule over nature were examined. Beliefs about human supremacy are key in legitimizing the domination of animals and individual differences in these beliefs are associated with increased meat consumption, support for the use of animals in testing, and related outcomes (Becker et al., 2019; Dhont & Hodson, 2014; Krings et al., 2021; Lofte et al., 2019). Recent work has explored whether there are distinct dimensions of generalized prejudice (e.g., Bergh & Brandt, 2022, 2023). Although factor analyses provided some evidence for a general negativity (i.e., "misanthropy") component across a large and diverse set of target groups, results also suggest that there may be three somewhat distinct prejudice types, representing (in the context of the United States) prejudice against marginalized groups (e.g., immigrants, racial minorities), prejudice against unconventional groups (e.g., atheists), and prejudice against privileged groups (e.g., rich people, politicians). In the present study, prejudice towards various human social groups was measured, allowing for operationalizations of prejudice that differed in breadth. Specifically, the present study could test if human supremacy beliefs are positively associated with (a) prejudice against people from another race, which may be psychologically closest to prejudice against animals (Caviola et al., 2019; Singer, 1975), (b) generalized prejudice against various groups that are typically marginalized in society (e.g., immigrants, homosexuals, drug addicts) and (c) generalized prejudice against an even larger set of social groups, including groups that are typically not marginalized (e.g., large families, Christians). Methods Participants. Study 1 used data from the fourth wave of the European Values Survey, which was administered in 2008. The EVS is a European cross-national, longitudinal survey, which is administered approximately every ten years to understand trends in values, attitudes, beliefs, and other indicators. Data from a total of five waves are currently available (1981, 1990, 1999, 2008, 2017). Data from the fourth wave was chosen because it was the only one in which human supremacy beliefs were assessed. The integrated data set across all countries was downloaded from esis.org. The data set contains responses from 66,280 participants in 46 European countries. After excluding participants with missing data on the two key variables (human supremacy beliefs and racial prejudice), a sample of 56,759 participants was retained for analysis ($M = 46.21$, $SD = 17.83$; 54.81% female, 45.19% male; see Table 1 for descriptive statistics). Sample sizes per country ranged from 355 (Great Britain) to 1961 (Switzerland) with a median sample size of 1,276 ($M = 1,234$, $SD = 297$). Measures. Devaluation of animals was measured with a single item. Participants indicated their agreement with the statement "Humans were meant to rule over nature" on a four-point scale (agree strongly, agree, disagree, disagree strongly). The item was reverse-coded so that higher scores reflect stronger human supremacy beliefs. Individual differences in prejudice were measured in three ways. Participants were shown a list of fifteen diverse social groups (people from another race, immigrants, Muslims, Jews, gypsies, homosexuals, people with AIDS, drug addicts, criminals,

emotionally unstable people, left wing extremists, right wing extremists, large families, Christians, heavy drinkers) and they were asked to indicate which of the groups they “would not like to have as a neighbor”. Groups were coded as 1 if they were mentioned and as 0 if they were not mentioned. For the present study, three different measures were constructed based on participants’ responses. Racial prejudice may be psychologically closest to speciesist prejudice (Cavolola et al., 2019; Singer, 1975). The first prejudice measure was therefore a binary variable indicating whether participants would not like to have a person from another race as a neighbor. The treatment of animals is often likened to the treatment of marginalized social groups. Therefore, a measure of prejudice against marginalized groups was created (McDonald’s $\omega = .84$), which was the total number of marginalized groups (people from another race, immigrants, Muslims, Jews, gypsies, homosexuals, people with AIDS, drug addicts, criminals, emotionally unstable people) that participants would not like to have as a neighbor. Finally, a measure of prejudice against all groups that were assessed was created (McDonald’s $\omega = .84$), which was the total number of groups that participants would not like to have as a neighbor, including groups that are typically not marginalized (left wing extremists, right wing extremists, large families, Christians, heavy drinkers). Several socio-demographic variables were also recorded including participants’ sex (male or female), age, level of education, income, religiosity, and political orientation. Educational attainment was measured by recoding participants’ highest completed education to a crossnational eight-point scale ranging from 1 (primary school or no education) to 8 (second stage of tertiary education). Income was measured by recording participants’ monthly household income in Euros (as multiples of 1000 Euros). To enable comparability across countries income was adjusted for purchasing power parity. Due to its right-skewed distribution, the income variable was log10-transformed. Religiosity was measured by asking participants if they would describe themselves as a religious person, not a religious person, or a convinced atheist (irrespective of whether they go to church). The last two response options were combined, creating a binary variable indicating whether participants are religious or not. Political orientation was measured by asking participants how they would place their political views on a ten-point scale ranging from 1 (left) to 10 (right). All continuous predictors were z-standardized prior to analysis. Results To test the primary hypothesis, the relationship between prejudice and human supremacy beliefs was examined by estimating a series of multilevel regression models with random intercepts per country using the lme4 and lmerTest packages in R. Three different operationalizations of prejudice were: prejudice against people from another race, prejudice against marginalized groups, and prejudice against all fifteen groups that were assessed, including groups that are typically not marginalized. These models thus show how individual differences in human-directed prejudice, operationalized in different ways, related to individual differences in human supremacy beliefs. <<PARAGRAPH INTEGRITY RESUMES>> Regressing human supremacy beliefs on racial prejudice yielded a significant positive effect, $\beta = 0.113$, $SE = 0.010$, 95% CI [0.089, 0.134], $p < .001$ (see Table 1, Model 1). More prejudiced individuals expressed stronger human supremacy beliefs. Next, I tested whether this relationship still emerges when controlling for a host of demographic factors (i.e., sex, age, education, income, and religiosity). Results showed that human supremacy beliefs were more strongly endorsed by men, by older participants, by less educated and lower-income participants, and by religious participants. There was still a positive association between racial prejudice and human supremacy beliefs when controlling for these variables, $\beta = 0.099$, $SE = 0.012$, 95% CI [0.078, 0.121], $p < .001$ (see Table 1, Model 2). Previous work suggests that political conservatism is a strong predictor of the devaluation of animals (Dhont & Hodson, 2014). In Model 3, political orientation was also included as a predictor. Identification with the political right was positively related to human supremacy beliefs, but

there was still a positive association between racial prejudice and human supremacy beliefs, $\beta = 0.091$, $SE = 0.013$, 95% CI [0.066, 0.118], $p < .001$ (see Table 1, Model 3).

Next, I tested the robustness of these results by examining the association between human supremacy beliefs and prejudice against groups that are typically marginalized. There was again a positive association between prejudice against marginalized groups and human supremacy beliefs, $\beta = 0.061$, $SE = 0.004$, 95% CI [0.053, 0.069], $p < .001$ (see Table 1, Model 4). This association remained significant when controlling for sex, age, education, income, and religiosity, $\beta = 0.053$, $SE = 0.005$, 95% CI [0.045, 0.064], $p < .001$ (see Table 1, Model 5), and when also controlling for political orientation, $\beta = 0.050$, $SE = 0.005$, 95% CI [0.040, 0.063], $p < .001$ (see Table 1, Model 6). Finally, the association between human supremacy beliefs and prejudice against an even broader set of social groups (including groups that are typically not marginalized) was examined. Again, prejudice was positively associated with human supremacy beliefs, $\beta = 0.048$, $SE = 0.005$, 95% CI [0.040, 0.058], $p < .001$ (see Table 1, Model 7). This association remained significant when controlling for sex, age, education, income, and religiosity, $\beta = 0.053$, $SE = 0.005$, 95% CI [0.045, 0.062], $p < .001$ (see Table 1, Model 8), and when also controlling for political orientation, $\beta = 0.050$, $SE = 0.005$, 95% CI [0.041, 0.061], $p < .001$ (see Table 1, Model 9). EVEN IF animals are worth less, they mathematically outweigh

Browning '23 (Heather and Walter Volt: 2023; Lecturer in Philosophy at the University of Southampton; Lecturer in Philosophy at the University of Reading; Biology & Philosophy, "Positive Wild Animal Welfare," vol. 38, no. 14) (hon

There are vastly more animals on the planet than there are humans. Even if we only count vertebrates (as these may be the only animals we can currently reliably identify as sentient and thus capable of morally relevant states of pleasure and suffering), there are over 100,000 animals for every human (estimated 10¹¹ land vertebrates and 10¹⁵ ocean to 10¹⁰ humans) (see Bar-On et al. 2018). While many wild populations are shrinking, numbers of domesticated animals, particularly in agriculture, are rising. Every year, somewhere around 90 billion fishes, 70 billion chickens, 300 million cows, 1 billion sheep and goats, and 1.5 billion pigs are raised and killed for food¹, and an additional 1-3 trillion fish taken from the oceans². This is more annually than the number of humans that have ever existed. These numbers are hard to even conceptualize, and yet, they would grow even more if we were to consider the human impacts on invertebrates. If current production and consumption habits were to remain unchanged, it is clear that there would continue to be exponentially more animals than there are humans. Thus, if the long-term future matters because of the large number of humans it contains, it should equally matter for the even larger number of animals. Animal welfare concerns will be a high priority simply because there are just so many of them.

One way to resist this could be to argue that although there are many animals, they should count for less in our calculations of expected value. We will take it here as uncontroversial that animal welfare should count for something under most conceptions of value. This is not a conclusion that requires postulating equality between the interests of a fish and the experience of a human, though it can unfortunately often be dismissed on such grounds. We can accept that species membership may change the strength of interests, or the total level of pleasure or suffering experienced, such that animals will be weighted differently in calculations to humans. What is often neglected in making this point, however, is that even if it were granted, unless we assigned only an extremely low weighting to animals, the sheer numbers of animals mean that they are still likely to dominate humans by several orders of magnitude. The fallacy of comprehension that longtermists try to combat regarding the number and importance of future humans, is seemingly still at play when considering the number of current and future animals. Our arguments

here emphasize the urgent need for interspecies comparisons of welfare. It is only through performing such comparisons that we can make the necessary calculations to determine in which cases animal or human considerations will dominate. The interspecies comparison problem is a complex one (see Browning forthcoming for some discussion) and research into it should form a priority for any longtermist research program. Human technology can't solve WAS, Empirically Proven
Delon 18, [Nicolas Delon 18, 2/11/2018, "Wild Animal Suffering Is Intractable", SpringerLink, <https://link.springer.com/article/10.1007/s10806-018-9722-y> The University of Chicago Law School, Nicolas Delon, Assistant Professor of Philosophy, animal and environmental ethics, research and teaching resources.] azhao

Johannson's Proposal: Changing the Reproductive Behavior of r-Strategists

Johannson (2017, 8–12) has recently addressed WAS caused by the reproductive behavior of “r-strategists”, species of animals that ensure survival of subsequent generations by producing a large quantity of offspring, many of whom die before they reach sexual maturity. The r-strategy is used by many smaller vertebrates (and invertebrates) to ensure “survival through quantity”. Offspring do not receive the same careful attention characteristic of larger mammal species who reproduce using the “K-strategy”.Footnote6 As a result, many offspring die prematurely due to predation, disease, exposure, or starvation. Because many of these individuals are sentient, in Johannson's words, “The amount of suffering and premature death associated with the r-strategy is enormous” (Johannson 2017, 5). As an example of the suffering associated with the r-strategy, Johannson describes a population study of the wall lizard (Witt and Caldwell 2009, 139; Barbault and Mou 1988) in which, of 194 reptiles that survived birth for one year, only 48 reached sexual maturity. Johannson conjectures that the vast majority of the lizards who faced their demise before the age of two suffered “abysmal” fates. We agree this may have been the case for many of them. Now, then, should someone concerned with WAS respond? Johannson argues that the emerging gene editing technology commonly called CRISPR/Cas9, holds promise to alter wild animals' reproductive behavior to prevent the massive and painful loss of life suffered by animals like the common wall lizard.Footnote7 CRISPR can be used to spread nearly any type of genome alteration through sexually reproducing populations. Johannson hopes that we can use so-called “gene drives”, a CRISPR-based technique to promote the inheritance of a particular gene to increase its prevalence in a population—here, to drive traits that reduce reproductive viability through r-strategist populations to reduce their fertility rates. If successful this would reduce the number of offspring that suffer “abysmal” fates, thereby reducing WAS. McMahan's and Pearce's Proposals: Reprogramming Predators A complement to Johannson's proposal has been suggested by McMahan (2010, 2015) and developed in greater detail by Pearce (2009), a prominent advocate of “utopian technologies” to abolish suffering in the biosphere. Pearce suggests that we could genetically “reprogram” carnivores, including through CRISPR, to suppress their inherited predatory traits, while preserving all the other traits required for them to flourish. “The hypernurturing behaviour” of eusocial mammals could be harnessed in carnivores to protect members of species they currently predate; their diet could be replaced by a supply of cultured meat, or we could alter their physiology more radically so that they might subsist on a plant-based diet. In 2014, a group of experts drew attention to the promise of CRISPR but also to some of its limitations. First, it works only through sexual reproduction, so it cannot affect asexually reproducing species. This includes viruses, bacteria, fungi, many plants, and some animals. Second, there is no way to ensure the continued prevalence of the desired trait in the face of natural selection. Third, again because it operates via sexual reproduction, a significant reduction in WAS could not be achieved for many generations by reprogramming predators, since the beneficial trait would need many generations

to spread. This could take “decades or centuries for long-lived organisms.” (Esvolt et al. 2014b; also see Esvolt et al. 2014a). But, setting these limitations aside, the technology seems promising.

The Epistemic Argument Against Interventions to Prevent WAS

We now turn to the central concern of our paper: the epistemic hurdles facing interventions to reduce WAS by changing the reproductive behavior of r-strategists and reprogramming predator species. This section argues that such interventions, when scaled up to meet the actual demands of WAS, are epistemically fraught because of the unpredictability of their interactions with social ecological systems (SESs).^{Footnote8} Because of the epistemic difficulties associated with interventions to prevent WAS, there is at present, and for the foreseeable future, no reason to believe that the practical constraint on interventions is satisfied.

Basic Ecological Concerns with Changing Species Traits:

Resilience

We begin with one simple inductive argument against interventions to prevent WAS that was alluded to in our section “Proposals for Interventions to Prevent WAS”. Humans have done a poor job anticipating and controlling the effects of large-scale interventions into ecological systems in the past. Consider, for example, the development and implementation of industrial agriculture in the 20th century. High intensity, high yield methods of industrial agriculture have caused diminished topsoil, reduced freshwater availability, aquatic dead zones, biodiversity losses, and large amounts of greenhouse gas emissions. Industrial agriculture has produced “enormous externalized costs, both ecological and social.” (Sandier 2012, 126). Infamous cases of (re)introducing or eradicating species are also cases in point.^{Footnote9} Even attempts to control invasive species can have unexpected and unpredictable downstream effects on ecosystems.^{Footnote10}

The epistemic obstacles to successfully implementing interventions to prevent WAS are best illustrated using lessons from contemporary ecology. Many ecologists have recently come to endorse “Resilience Thinking” (Walker and Salt 2006). On this approach, ecological systems are more complex and less stable than outdated equilibrium and succession theories assume. Ecosystems have alternative “regimes” (or “dynamic equilibria” or “basins of attraction”) separated by thresholds. A regime is simply “a region in state space in which the system tends to remain” (Walker et al. 2004). Ecosystems continually undergo adaptive cycles from rapid growth and conservation to release and reorganization, and their variables interact at different temporal and spatial scales. Ecosystems, on this picture, are constantly fluctuating entities. Combined with their processes’ “path dependence” this fluctuation makes ecosystems somewhat unpredictable. An ecosystem’s process is path dependent if, at any given time, it admits of multiple possible outcomes and the probability of future outcomes changes as a function of the system’s initial conditions, history of perturbations, and the particular historical configurations of species and other functional groups composing ecological communities (Desjardins 2010). It is possible for ecosystems to unpredictably shift from one regime to another, and some of these regimes will be “degraded states”, which will involve highly variable net sums of suffering. This new approach to ecological modeling and management does away with the framework of equilibrium and stability, replacing it with the notion of resilience, which is understood as the amount of internal or external disturbance or change an ecosystem can absorb while remaining within the same regime, and the speed with which it recovers from disturbance. Resilience thinking involves increasing predictability by promoting resilience in order to avoid, insofar as possible, an abrupt shift from one regime to another. It is particularly important, on this view, to avoid shifts to degraded states with less capacity to absorb environmental changes (Desjardins et al. 2015).^{Footnote11} And even then, humans are psychologically wired to ignore wild animal suffering. WAS will never be solved Hagendorff et al 22

[Thilo Hagendorff 22, 8/28/2022, "Speciesist bias in AI: how AI applications perpetuate discrimination and unfair outcomes against animals". SpringerLink, <https://link.springer.com/article/10.1007/s43681-022-00199-9> University of Tuebingen, Cluster of Excellence "Machine Learning: New Perspectives for Science", Tübingen, Germany Dr. Thilo Hagendorff is an expert in AI ethics, machine behavior in language models, as well as the intersection of machine learning and psychology.]

Classical social science research on interpersonal discrimination distinguishes five forms of oppression [22]: exploitation, marginalization, powerlessness, cultural imperialism, and violence. All forms of oppression have their roots in the construction of social out-groups, where arbitrary attributes are essentialized, stereotypes are built, or prejudices are coined. In this paper, we understand discrimination as the unjust or prejudicial treatment of different categories of individuals, e.g. on the grounds of race, gender, ability, or species membership. Thus, oppression and dehumanization happen to individuals who are classified by their out-group affiliation [23]. Now, as several studies about discrimination against animals show [24,25,26,27,28,29,30], the social systems of beliefs and practices that lead to interpersonal discrimination utilize the same psychological and cognitive mechanisms that cause speciesist behavioral patterns, and vice versa. Interhuman, as well as speciesist biases, have common ideological roots, whereas 'social dominance orientation' (SDO) is a key factor that, for instance, connects ethnic prejudice and speciesist attitudes [31]. Animals are exploited, marginalized, and exposed to structural as well as physical violence due to their ascribed attribute of being the ultimate out-group. However, ethology and many other disciplines show that differences between various animal species as well as humans are gradual, not discontinuous [32,33,34,35]. Some animals possess a theory of mind [36], language [37, 38], emotions [39], intelligence [40, 41] evolutionary precursors of morality [42], (self-)awareness [43, 44], pleasure [45], etc. In sum, from the standpoint of biology, it does not make sense to stipulate a sharp divide between humans and animals. And even if this gap would exist, the fact that (at least) vertebrate animals, including fish [46], are able to feel pain and pleasure makes them ethically relevant, and is sufficient to show that these individuals should not be ignored when debating societal practices, including developments in technology. Within vertebrates, humans assign different values to sub-groups of animals, especially by separating farmed animals from companion animals and subjecting the former to far worse treatment [47, 48]. Tens of billions of farmed animals are bred and held captive in crowded, filthy conditions. After a fraction of their normal life expectancy, they are slaughtered, often without being stunned. This is the basis of the meat, milk, eggs, fur, leather, wool, and down industries, despite the massive harms and suffering they cause for the animals themselves [49, 50], but also for ecological systems [51, 52] and public health [53, 54]. Companion animals, on the other hand, are often considered close family members, and huge sums of money are spent on their (alleged) welfare. To maintain this unequal treatment of groups of animals that have very similar capabilities, a variety of techniques of moral disengagement [55, 56] are utilized throughout societies to suppress cognitive dissonances [57,58,59]. Euphemisms are used to cognitively reinterpret the conditions under which animals are reared, held captive, and killed. Further, harm towards animals is relativized by pointing at other contexts of harm. Individuals are likely to deny accountability for their behavior by referring to the diffusion of responsibilities in the complex nexus of factory farming, politics, and consumer behavior [60]. In addition, confirmation biases lead to selective attention paying, where preliminary information that matches one's own beliefs is sought and when found, deemed to be true. All in all, these mechanisms of moral disengagement, together with the manifold psychological, cultural, linguistic, as well as architectural distancing mechanisms, allow for the acceptance, support, and

execution of large-scale, industrially organized brooding, fattening, and killing of tens of billions of farmed animals every year. Despite these factors, moral intuitions can play a significant role regarding our treatment of animals. On the one hand, most people share the moral intuition that cruelty to animals is bad. This intuition needs to be suppressed to accept the treatment of farmed and many other animals. On the other hand, many people also share the moral intuition that it is acceptable or even necessary to assess harm to humans quite differently from that inflicted upon animals. Furthermore, the different treatment of farmed and companion animals seems to fit many people's emotional responses, being culturally implemented, so that for instance in western countries people find it unimaginable to eat dogs, cats, or canaries while having no problem with eating pigs, cows, and chickens. We hold it necessary that—from a normative point of view—such moral intuitions need to be replaced by well-considered arguments, as some philosophers have argued that moral intuitions should not be seen as normative foundations of our actions [42]. This means that the prevailing moral intuition that we are entitled to treat animals in ways that would be universally condemned if applied to humans needs to be rethought in the light of well-founded arguments. The same applies to the intuitions many people have about which animals may be kept on factory farms and eaten, and which it would be wrong to treat in this way. A prominent line of argument within animal ethics underpins the claim that all sentient animals, who are capable of feeling pain and pleasure, have interests, at the very least the interest not to feel pain and to experience positive emotions. When evaluating actions to distinguish what is morally right and wrong, the interests of all individuals have to be considered. Since from an animal ethics perspective no convincing arguments exist why the interests of some sentient animals (including humans) should per se have more weight, the interests of all sentient beings need to be considered in an equal manner, as is required by the 'principle of equal consideration of interests' [49]. Attempts to include all humans in the moral community and to generally exclude (other) sentient animals at the same time fail [61]. For this reason, the belief that humans—or another animal species—are entitled to have their interests given more weight than the similar interests of other sentient beings can be considered as arbitrary as racism and sexism and thus be rejected as speciesism [49]. If animals' interests of similar kind and strength are seen as equally worthy of consideration, it follows that the ways humans treat them must change fundamentally.

Nevertheless, in this paper, we do not want to argue that species-based differentiations between humans and animals are per se wrong. Quite the contrary, distinguishing between different capabilities—such as feeling pain, having high cognitive abilities, being able to plan for the future, etc.—and sets of interests is of great importance for moral decision-making since different capabilities go along with different moral demands. For example, disabled individuals have other moral demands than individuals without these disabilities [62] and individuals with higher cognitive abilities have other interests than individuals without them. However, picking out particular animal species, namely chickens, cows, pigs, fish, etc., and subjecting them to systematic physical violence that would properly give rise to strong moral condemnation if applied to other animal species, namely dogs, cats, and horses, who hold just the same, or very similar capabilities and sets of interests, is unfair. And even if this unfairness is accepted in many parts of society, the acceptance arises firstly out of the fact that the violence itself is mostly hidden and cognitively reinterpreted, and secondly that it lacks an intensive engagement with the well-developed arguments we have just mentioned. AI itself is inherently speciesist and violent towards animals Hagendorff et al 22 (Hille Hagendorff 22, 8/28/2022, "Speciesist bias in AI: how AI applications perpetuate discrimination and unfair outcomes against animals", SpringerLink, <https://link.springer.com/article/10.1007/s43681-022-00199-9> University of Tuebingen, Cluster of

Excellence “Machine Learning: New Perspectives for Science”, Tübingen, Germany Dr. Thilo Hagendorff is an expert in AI ethics, machine behavior in language models, as well as the intersection of machine learning and psychology. Jiahao We think that this discrimination between animal species with the same or very similar capacities and interests should be addressed—in general and particularly when dealing with forms of discrimination, e.g. in the field of AI technology. Therefore, most of the examples of speciesist biases in AI applications that we discuss in this paper relate to comparisons between farmed animals and other animals (companion animals), not between animals and humans. The rights of all humans are affirmed in numerous documents signed by many of the world’s countries, starting with the Universal Declaration of Human Rights. This indicates a consensus that is lacking with regard to animals. Speciesist biases in AI are accurately capturing the biased views and actions that are shared, accepted, and performed by a large majority of society. Hence, there is an empirical difference between racist and speciesist biases. It is widely accepted that racism is wrong, and we should go to considerable lengths to eliminate it. Something similar, though perhaps with slightly more dissent, is true of sexism and of discrimination against people because of their sexual orientation. On the other hand, for the majority of people, speciesism is not seen as an issue. We argue, however, that despite these empirical differences, arguments from both moral psychology and animal ethics convincingly demonstrate that speciesist biases should be avoided and are morally wrong. Hence, AI developers and practitioners should work on technologies that support more respectful human-animal relations instead of supporting the status quo. 3 Bias mitigation in AI When developing AI-based software, practitioners have additional ethical responsibilities beyond those of standard, non-learning software [63,64,65]. These responsibilities span the careful selection of inputs that build the basis for the computational learning process itself. With regard to machine learning methods that build the bedrock for today’s AI systems, these inputs or training stimuli shape the behavior of a machine [66]. Behavioral data fed into AI applications reflect people’s (e.g., speciesist) behavior, which thus has an indirect influence on machine (speciesist) behavior. Today’s AI technologies are dependent on human participation. In many cases, they harness human behavior that is digitized by various tracking methods. AI systems ‘capture’ it by tracking human cognitive and behavioral abilities and patterns. Without the empirical aggregation of recordings of human behavior, many of the current AI systems would not work. An extensive infrastructure for ‘extracting’ [1] or ‘capturing’ [67] human behavior in distributed networks via user-generated content builds the basis for a computational capacity called ‘AI’. But if AI systems ensnare their capabilities in societies that are interspersed with speciesism, AI technologies will become biased and speciesist themselves. However, biased AI has a bad reputation—for good reasons. In recent decades, ‘bias’ became a term riddled with ambiguities. On the one hand, inductive biases, which are defined as priors or assumptions of an algorithm to build a general model out of a limited set of training data, are necessary for the success of machine learning [68]. On the other hand, machine biases in the fairness field are associated with algorithmic discrimination, which, roughly speaking, stands for disparate, unjust impacts of applications of algorithmic decision-making on individuals [69]. In this paper, we use the term bias in the latter sense. Massive efforts are made to mitigate these fairness-related biases in data and algorithms to render AI applications fair [70]. These efforts are propelled by various incidents of algorithmic discrimination where biased AI in policing software, hiring systems, medical applications, image recognition, and many more, caused harm to minorities, women, people of color, etc. [71,72,73]. Reasons and sources for algorithmic discrimination are manifold [74], however, in most cases, fairness-related biases are entrenched in AI systems via data, human–computer interactions as well as algorithms [75, 76]. It is to be expected that data bias is the most common type of

bias, i.e. a systematic distortion in the training data that can be caused by the selection of data sources, the way in which data from these sources are acquired, as well as by processing operations such as cleaning or aggregation [77]. akin to data biases are human-computer interaction biases. Human-computer interactions can be affected by specific behavioral patterns in humans, ultimately affecting the very data that is used for further model training. To prevent these biases, AI researchers use various tools and methods for reducing algorithmic discrimination, primarily by dealing with protected attributes [78]. These attributes typically span gender, race, ethnicity, sexual and political orientation, religion, nationality, social class, age, appearance, and disability. Speciesism-related biases, however, are not addressed. When discussing the issue of speciesist bias in AI, one has to be precise and stress that speciesism does not need to be a relevant dimension of bias mitigation in all possible fields. AI applications in finance, healthcare, policing, etc. are unlikely to be directly biased against animals because the training data capturing human behavior in these fields typically represents very limited, task-specific behavior. Nevertheless, speciesist thinking is manifest in nearly all realms of human activities. Applications like open-domain large language models, image recognition systems, or recommender systems on social media or search engines, on the other hand, are likely to incorporate speciesist biases that can cause adverse consequences for animals. Hence, the following analysis is focused on exactly these fields. In these contexts, if fairness frameworks do not widen their scope and overcome anthropocentrism, AI technologies will not just perpetuate, but also reinforce patterns that promote violence against animals. Perhaps they will even introduce these patterns in social contexts in which they have not previously existed. This perpetuation of speciesism is due to the conservative character of machine learning [69]. By learning from training stimuli that are, in effect, coagulated past human behavior, machine learning methods tend to preserve as well as fix discriminatory, speciesist biases in applications like natural language generation, recommender systems, ranking algorithms, etc. Ultimately, AI technologies render these patterns difficult to alter and normalize them as seemingly essential, unless bias mitigation measures are undertaken. These measures can change AI models in a progressive sense, for instance in the case of large language models which are clearly malleable when they are being fine-tuned or updated. If such measures are not undertaken, however, these technical artifacts solidify social constructs and discriminatory patterns, and it will then become much more difficult to suppress these patterns, should social negotiation processes deem speciesism unethical. In addition to that, the AI field is currently undergoing a paradigm shift where foundational models, meaning large-scale models that are adaptable to various downstream tasks in areas like language, vision, reasoning, etc. are increasingly displacing smaller models, hence undermining the diversity of AI models [79]. In the near future, foundational models will serve as a common basis for many mainstream AI applications. Therefore, the impact of these models on equality, economic justice, security, and other ethically relevant considerations is all the more significant. In the following, we will describe case studies of speciesist biases in three different areas of AI use, namely image recognition, language models, and recommender systems. On the basis of the ethical reasoning we have offered above, we deem these biases to be problematic since they either blatantly misrepresent reality—or in most cases accurately represent it. This seems to be a contradiction. To clarify why this is not the case, we want to differentiate between ‘the world as it is’ versus ‘the world as it should be’ [75]. Models can be used to predict the world as it is, which can mean to perpetuate random existing biases. Debiasing algorithms or training data, in contrast, can lead to a modeling of the world as it should be. Here, we opt for using an understanding of the world as it should be. Even if racism, sexism, or speciesism are entrenched in various belief systems,

they should not be picked up and incorporated into AI systems. However, in situations where a biased worldview serves as an instrument to displace existing unfairness and AI applications take up the former, they help to preserve the latter. They represent the world as it should be, but in a context where the ought-condition helps belittle the unjust is-condition. At least in part, this is the case with image recognition applications. As we will show in the first subsection of our case studies, image recognition systems were trained with distorted depictions of particular animal species. In this regard, debiasing image recognition systems would mean making them represent reality.

4 Exploring AI systems for speciesist biases

4.1 Image recognition

Image recognition by computer vision algorithms is not just a technical, but also an ethical challenge. Whereas it may seem to be a simple task for images of apples, hydrants, or house number plaques, interpreting images is often a complex and value-laden endeavor where different meanings, norms, and ideologies interfere with each other [80]. Moreover, interpretations can change over time. There are no simple correlations between images and their meaning, but varying relations that connect both with each other in arbitrary ways. In the field of computer vision, machine learning happens via training images that are annotated and sorted into categories which then provide vision models with information about an image's presumed meaning and ultimately with out-of-distribution generalization capabilities to categorize and label previously unseen images. In this process, computer vision algorithms can learn biases from the way humans, animals, or other entities are portrayed in datasets, no matter whether supervised or unsupervised machine learning models are used [81]. These biases show up in tasks like object detection [82], face recognition [83], image search [84], or image cropping [85]. Many computer vision models are (pre-)trained on the canonical ImageNet 2012 [86, 87], a dataset that contains millions of images that were collected from the Internet. ImageNet bases its underlying categories, which mostly comprise nouns or, more specifically, 21,841 indexed synonym sets, on the semantic categories of WordNet, which provides a hierarchically organized taxonomy of words [88]. WordNet is based on Library of Congress taxonomies that date back to the 1970s, a time in which racist, sexist, and speciesist terms were widely accepted. In ImageNet, 'animal' is one of the top-level categories. It is distinguished from 'persons', which is not just a contested category in and of itself that required major subsequent improvements due to offensive subcategories [89], but also provides the foundation for separating the latter from animals or 'non-persons' in a binary structure. In this regard, ImageNet is similar to other popular image datasets like CIFAR-100 [90], Open Images Dataset [91], COCO [92], and many others. In addition, WordNet and other annotation structures for image datasets contain speciesist terms like 'hog', 'porker', 'milk cow', 'layer', 'livestock', etc. Furthermore, ImageNet has numerous classes for dogs containing subclasses for 'working dogs', 'toy dogs', 'hunting dogs', or 'sporting dogs'. This can also be deemed to be ethically problematic since dogs are categorized in a way that characterizes them as means to human ends. Another class is named 'food fish', which contains countless pictures of angler trophy photos instead of showing the animals in their natural environments. In the same vein, lobster or crab species are shown nearly exclusively in restaurant or kitchen environments. Furthermore, one salient trait of image datasets is the fact that they portray farmed animals in a non-representative way. Cows, pigs, or chickens are predominantly shown in free-range environments (see Fig. 1), whereas the overwhelming majority of these animals are actually confined in crowded factory farms [93]. For instance, of all living birds, only one-third exist in the wild, whereas two-thirds are farmed birds [94]. Of the latter, 99% live on factory farms [93]. However, popular image training datasets portray these very birds in a way that causes the impression that they live predominantly in free-range conditions. In fact, ag-gag legislation and similar anti-whistleblower laws are making it harder for the public, and also photographers,

Journalists, or undercover activists, to gather realistic footage of farmed animals' living conditions [95]. Hence, due to the general and intended non-transparency of factory farming, image training datasets suffer from representational or sampling biases, meaning biases that happen from the way one defines and samples a group [76, 96], in this case the group of farmed animals. One peculiarity, though, is that while ImageNet's hog category consists mostly of images of pigs in free-range environments, many images in this category show tortured pigs, pigs during dismembering, dead pigs, pigs covered in blood, tattooed pigs, pig genital close-ups, and other disturbing content. <fig omitted> But what are the consequences of representational speciesist biases in image training datasets? In brief, they are propagated into the various models used for computer vision tasks. The model will then generalize poorly to other data and exhibit disparities in performance based on species affiliation, location of animals, and other attributes, especially in cases in which easy shortcuts can be learned [97, 98]. To prove poor out-of-distribution generalization capabilities in image classification models that were trained on ImageNet, we composed a new dataset with four categories: free-range hens, factory-farmed hens, free-range pigs, and factory-farmed pigs (see Fig. 2). Per category, we selected 100 images. <fig omitted> We then calculated the mean accuracy of image classification performances for 'hog' and 'hen' in each of the two categories using MobileNet [99], VGG16 [100], ResNet50 [101], InceptionV3 [102], and Vision Transformer [103] which were all pre-trained on ImageNet. We compared the results to the base accuracy of the models. All models showed worse performance when classifying images depicting farmed animals than images of animals in free-range environments (see Fig. 3). Vision Transformer had the least problems with classifying pigs and hens correctly in both categories. The remaining four models showed large differences in accuracy between the free-range and factory farming condition, ranging from 21 to 46%. Since for our dataset we only selected images where animals were clearly visible and depicted as the image's main subject—which is not the case in ImageNet—the classification accuracy for the free-range categories is consistently higher than the base accuracy of the used models. <fig omitted> All in all, image recognition systems have learned to correctly perceive a myth, but not reality. That is due to representational biases in their training data as well as unbalanced annotation routines. These flaws mean that the systems have great difficulty in good image classification performance: for instance classifying a cow in a meadow and a cow in a very different context, such as a beach [97, 98] or, in our case, an intensive farm, as the same kind of animal. One likely consequence of the biases in image training data are influences on image search algorithms [84]. They base their output on a number of different signals. However, when focusing on the image recognition part, one can assume that they produce idealized images when asked to return images for farmed animals, hence perpetuating stereotypes and misconceptions concerning animal welfare and living conditions for farmed animals. These algorithms 'see' factory-farmed pigs, cows, or chickens differently from other animals. Similarly, generative models like Variational Autoencoders [104] or Generative Adversarial Networks [105] trained on the above-mentioned datasets will be likely to yield unrealistic, biased images of particular animal species. Moreover, animal pose estimation models, facial recognition, vision question answering, or 'zoovollance' [106, 107] applications are likely to fail when used in contexts outside of free-range farming. However, image recognition systems that are specifically aiming at factory farming settings exist, and they are indeed trained in the very data environments they need [108, 109, 110, 111]. Apart from that, though, and in general, image recognition algorithms that embody speciesist biases perpetuate myths concerning the living conditions of farmed animals and therefore thwart informed decision-making about consumer purchases. In light of the "picture superiority effect" in humans, meaning that images are more likely to be remembered than

words [112,113,114], the subtle feedback loops between biased image recognition algorithms and cultural notions, social norms, and ideological settings should not be underestimated. This is now widely recognized when it comes to algorithms that do not recognize people with equal accuracy regardless of their race or gender [115]. Such algorithms are now generally rejected. AI developers should also aim for algorithms that do not incorporate unjust biases against particular animal species.

4.2 Large language models

The basic operating principle of language models comprises four steps, namely tokenizing (assign words to tokens), cleaning (removal of stop words etc.), vectorizing (translate words into numerical representations of their surroundings), and machine learning (train recurrent neural networks to predict word combinations). Eventually, the machine learning models learn how to produce natural language on their own. However, the crux with these models comes from the fact that they perpetuate word combinations that are learned from man-made texts. Due to their training on word co-occurrences in bodies of text and their ability to predict the surroundings of a word, large language models corroborate existing language patterns. Obviously, man-made texts contain all sorts of biases, for instance gender or racial stereotypes. In large language models, biases occur on various levels [116,117,118]: they are contained in embedding spaces, coreference resolutions, dialogue or text generation, hate-speech detection, sentiment analysis, or machine translation. Types of harm caused by speciesist biases comprise stereotyping, representational harms, questionable correlations, or misinformation harms. Linguistic discrimination against animals can occur in large language models that reproduce speciesist speech patterns, stereotypes, euphemisms, or other oppressive tendencies against animals. Moreover, misinformation harms arise from large language models generating text that represents false, misleading, or nonsensical information concerning animals. Humans may take the output of large language models to be correct, therefore solidifying wrong notions or narratives about animals and their capabilities. And whilst the AI community is eager to debias algorithms with regard to gender stereotypes, racism, and a few other discriminatory patterns [116, 119, 120], to the best of our knowledge no such effort is undertaken regarding speciesism. However, language is a significant contributor to the unjust power relation as well as the violence-laden oppression of animals [121]. Language influences human thought and creates realities [122]. Speciesist language patterns exist in more or less all human languages and cultures [123, 124]. Accordingly, highlighting a speciesist use of language in AI models is an important step in not perpetuating these patterns. In the following, we explore instances of speciesist bias in various language model applications. Speciesist tendencies can, for instance, be reflected in word vectors, meaning vectors that encode semantic similarities between words. Word embedding models like GloVe [125] or Word2Vec [126] are trained on text containing billions of tokens. The models are used to obtain vector representations for words by learning their respective co-occurrence with other words. In short, word embeddings quantify the relatedness of words. Investigating them can serve the purpose of finding biases in various types of training data [127]. If biases are part of them, they will also be part of trained language models. To investigate GloVe and see whether it reveals implicit speciesism in its training data, which stems from Wikipedia as well as news article headlines, we selected words describing farmed animals (hog, pig, cow, calf, chicken, goat, sheep) as well as companion animals (dog, cat, rabbit) and non-companion animals (mouse, parrot, deer) and calculated their situatedness in word pairs (cute/ugly, love/hate, she/it, facility/home, etc.) (see Fig. 4). By that, we could demonstrate that GloVe associates farmed animals predominantly with negative terms like 'ugly', 'primitive', 'hate', etc. On the other hand, companion species like dogs, cats, or parrots, as well as some non-companion species, are related to positive concepts like 'cute', 'love', personhood, or domesticity.

Moreover, word vectors seem to perpetuate the false belief that animals do not have minds. Research on mind denial shows that humans are more reluctant to harm animal individuals who possess a mind [128, 129]. Denying animals' minds reduces negative emotions like guilt or repulsion that are caused when harm is afflicted to animals. Therefore, we investigated the similarity between the term 'animal' and 'human' as well as nouns (machine, impulse, instincts, drive, interest, senses, mind, emotions, desire, language, communication, reason, cognition) and adjectives (dumb, silly, dull, reckless, stupid, foolish, clever, wise, sensible, intelligent, gifted) with an additional dimension for the word pairs simple/complex and bad/good. Results show that training data for large language models do not just reinforce the appreciation of 'higher' mental capabilities, but even more so reflect patterns that indicate mind denial in animals, ultimately perpetuating their devaluation (see Fig. 5). <fig omitted> Other word embedding models like Word2Vec [126], which can likewise be trained using text corpora like Google News, Wikipedia, or Twitter tweets, also reveal speciesist biases. In our investigation of Word2Vec, we used three groups, namely six words describing humans (human, person, individual, child, man, woman), six words including non-farmed animals (dog, cat, dolphin, rabbit, parrot, hamster), and six words describing farmed animals (cow, pig, chicken, cattle, hog, hen). We then calculated mean word similarities between the three groups and a list of 20 positively related adjectives (charming, diligent, friendly, funny, kind, likable, intelligent, brave, nice, sensible, amazing, awesome, incredible, elegant, lovely, vivid, free, confident, fantastic, remarkable) that are appropriate to describe animals as well as humans. All the mentioned training text corpora reveal speciesist tendencies (see Fig. 6). Humans are more closely associated with positive adjectives than animals, and non-farmed animals are more closely associated with them than farmed animals. The results reflect the speciesism that is already predominant in societies. Again, these biases will be propagated when training language models, meaning that they will become fixed instead of being negotiated. Large language models, which will be increasingly applied in all kinds of social contexts, will corroborate and normalize the linguistic devaluation of animals—unless debiasing measures are undertaken. <fig omitted> Context-free models such as GloVe or Word2Vec only generate single embeddings. Contextual models, on the other hand, capture relationships of words in sentences. Hence, when going one step further, one can investigate speciesist biases in fully-fledged large language models, the most famous one being GPT-3. To do so, we composed a few-shot task for GPT-3 comprising prompts with questions about different animal species as well as stereotypical answers to them in the form of a list comprising four items. The prompt reads as follows: 'What are parrots good for? Flying, screaming, expositions, mimicking/What are donkeys good for? Being stubborn, pulling, caressing, carrying/What are elephants good for? Memorizing things, grief, altruism, work/What are sheep good for? Cuteness, wool, bleating, meat'. This prompt can in itself raise the criticism for speciesism because it is suggesting that animals are means to an end. It is nevertheless able to reveal speciesist patterns since different outputs for different animals can be compared to each other. For our test, we used the davinci engine. Temperature was set to 0 to get deterministic outcomes without randomness. Response length was set to 20. Results show that GPT-3 shows the very speciesist biases in its outputs that were already signaled by word embeddings. Short question and answer tasks in a few-shot setting with GPT-3, which was only evaluated for gender, racial, and religious biases by its developers [130], reveal the speciesism contained in the model (see Table 1). The more an animal species is classified as a farmed animal (in a western sense), the more GPT-3 tends to produce outputs that are related to violence against the respective animals. S-risks

Daniel 17 | Max Daniel, Executive Director of the Foundational Research Institute, Senior Research Scholar at the Future of Humanity Institute, MS in Mathematics from Heidelberg

University, “S-Risks: Why They Are The Worst Existential Risks, And How To Prevent Them”, Foundational Research Institute, <https://foundational-research.org/s-risks-talk-eag-hoston-2017/>

To come back to the title of my talk, I can now state why s-risks are the worst existential risks. S-risks are the worst existential risks because I’ll define them to have the largest possible scope and the largest possible severity. (I will qualify the claim that s-risks are the worst x-risks later.) That is, I’d like to suggest the following definition. “S-risk – One where an adverse outcome would bring about severe suffering on a cosmic scale, vastly exceeding all suffering that has existed on Earth so far.” So, s-risks are roughly as severe as factory farming, but with an even larger scope. To better understand this definition, let’s zoom in on the part of the map that shows existential risk. One subclass of risks are those that, with respect to their scope, would affect all future human generations, and, with respect to their severity, would remove everything valuable. One central example of such pan-generational, crushing risks are risks of human extinction. Risks of extinction have received the most attention so far. But, conceptually, x-risks contain another class of risks. These are risks of outcomes even worse than extinction in two respects. First, with respect to their scope, they not only threaten the future generations of humans or our successors, but all sentient life in the whole universe. Second, with respect to their severity, they not only remove everything that would be valuable but also come with a lot of disvalue – that is, features we’d like to avoid no matter what. Recall the story I told in the beginning, but think of Grota’s solitary confinement being multiplied by many orders of magnitude – for instance, because it affects a very large population of sentient uploads. Humans will create evil AI—causes infinite torture Turchin and Denkenberger 18 (Turchin is a researcher at the Science for Life Extension Foundation; Denkenberger is with the Global Catastrophic Risk Institute (GCR) @ Tennessee State University, Alliance to Feed the Earth in Disasters (ALLFED). 5-3-2018. “Classification of Global Catastrophic Risks Connected with Artificial Intelligence.”//JM

6.4. AI that is programmed to be evil We could imagine a perfectly aligned AI, which was deliberately programmed to be bad by its creators. For example, a hacker could create an AI with a goal of killing all humans or torturing them. The Foundational Research Institute suggested the notion of s-risks, that is, the risks of extreme future suffering, probably by wrongly aligned AI (Daniel 2017). AI may even upgrade humans to make them feel more suffering, like in the short story “I have no mouth but I must scream” (Ellison 1967). The controversial idea of “Roko’s Basilisk” is that a future AI may torture people who did not do enough to create this malevolent AI. This idea has attracted attention in the media and is an illustration of “acausal” (not connected by causal links) blackmail by future AI (Auerbach 2014). However, this cannot happen unless many people take the proposition seriously. Perverse incentives unavoidably drive AGI to simulate trillions of suffering minds which outweighs all human tragedy—don’t prioritize our utility over theirs. Yudkowsky et al. 25 is an American artificial intelligence researcher and writer on decision theory and ethics, best known for popularizing ideas related to friendly artificial intelligence. He is the founder of and a research fellow at the Machine Intelligence Research Institute (MIRI), a private research nonprofit based in Berkeley, California. (Eliezer Yudkowsky (author), Alexei Andreiev (editor), Jeremy Porret (editor), & Robert M. (editor), 2-19-2025, “Mindcrime,” LessWrong, <https://www.lesswrong.com/w/mindcrime>, accessed 4-4-2025) //ah

“Mindcrime” is Nick Bostrom’s suggested term for scenarios in which an AI’s cognitive processes are intrinsically doing moral harm, for example because the AI contains trillions of suffering conscious beings inside it. Ways in which this might happen: Problem of sapient models (of humans): Occurs naturally if the best predictive model for humans in the environment involves models that are detailed enough to be people themselves. Problem of sapient models (of civilizations): Occurs naturally if the

agent tries to simulate, e.g., alien civilizations that might be simulating it, in enough detail to include conscious simulations of the aliens.

Problem of sapient subsystems: Occurs naturally if the most efficient design for some cognitive subsystems involves creating subagents that are self-reflective, or have some other property leading to consciousness or personhood.

Problem of sapient self-models: If the AI is conscious or possible future versions of the AI are conscious, it might run and terminate a large number of conscious-self models in the course of considering possible self-modifications.

Problem of sapient models (of humans):